

When **Similar Places** Fool AI

Which satellite tile is the same place?

UAV QUERY



A



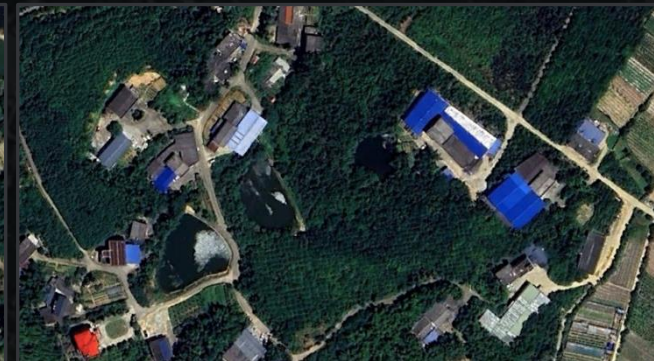
B



SATELLITE MAP CANDIDATES



C



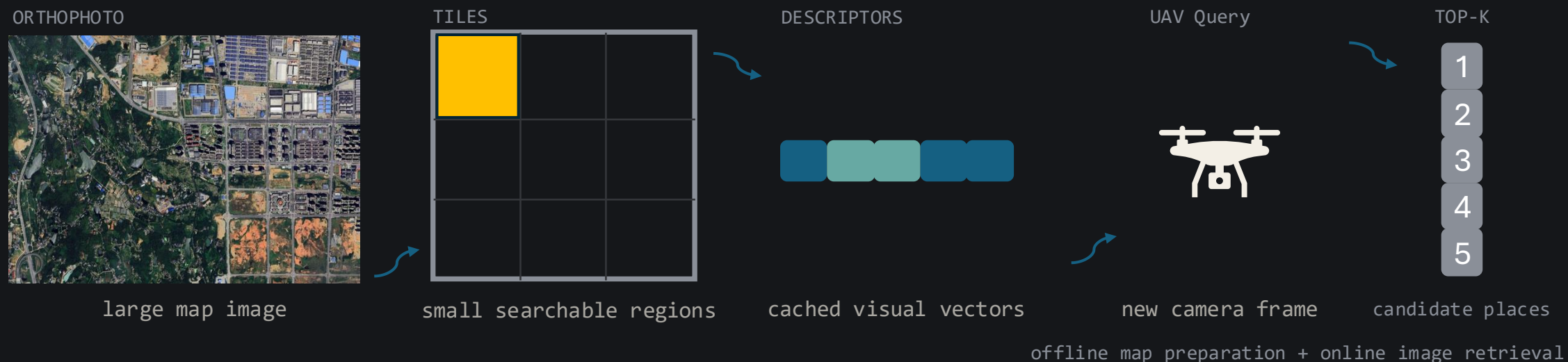
D

Which one would you **trust**?

NO GPS • NO COORDINATES • IMAGE ONLY

The map becomes **searchable memory**

We do not search GPS coordinates — we search **visual descriptors**.



Reference coordinates are used **only after retrieval** — for evaluation.

The right answer must enter the **candidate pool**.

MAP TILES → DESCRIPTORS → RETRIEVAL

Why the **same place** looks different

*UAV-to-satellite matching is **not just image similarity**.*

VEGETATION / MAP AGE



SHADOWS



CONSTRUCTION CHANGE



WATER BODIES



REPEATED STRUCTURE



ORIENTATION

Visual similarity is not geographic identity.

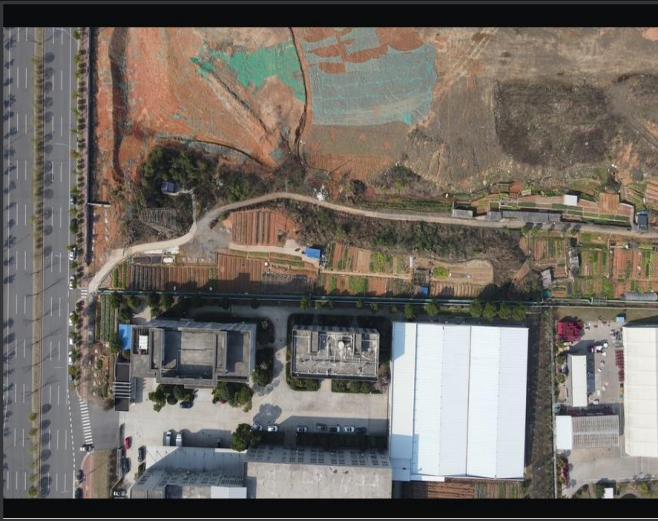
FAILURE MODES BEFORE METHOD CHOICE

Classical methods found a **believable wrong place**

The wrong tile was not random — it shared structure, edges, and visual layout.

UAV QUERY

TOKEN 67



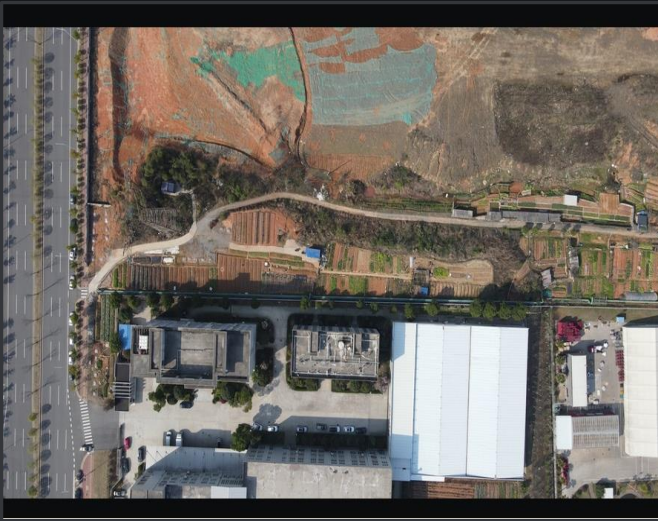
Where would classical methods look first?

Classical methods found a **believable wrong place**

The wrong tile was not random — it shared structure, edges, and visual layout.

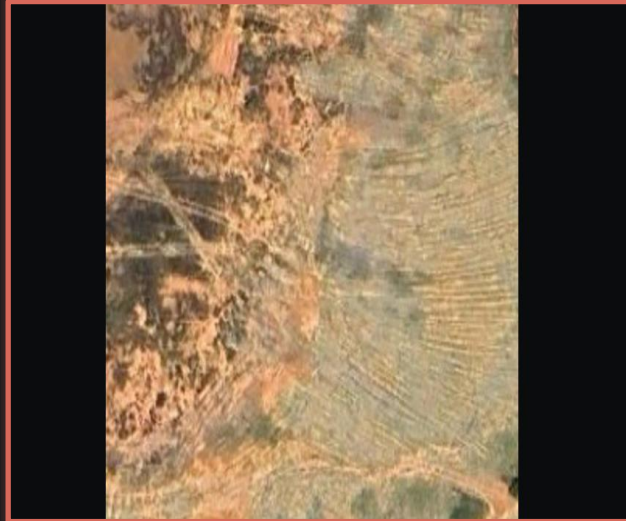
UAV QUERY

TOKEN 67



CLASSICAL TOP-1

TOKEN 67



Wrong — but visually plausible

Classical methods found a **believable wrong place**

The wrong tile was not random — it shared structure, edges, and visual layout.

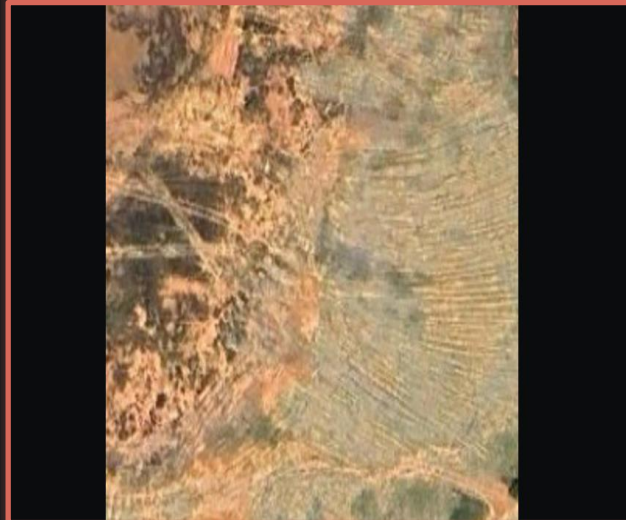
UAV QUERY

TOKEN 67



CLASSICAL TOP-1

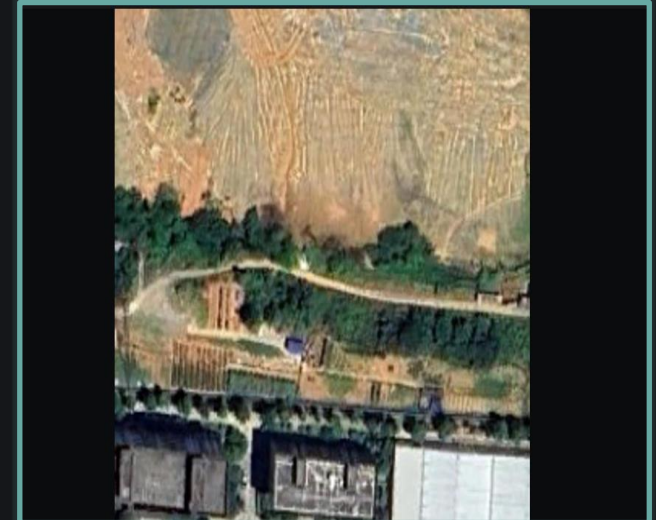
TOKEN 67



wrong — but visually plausible

DINOv2-VLAD

TOKEN 67



recovered the place

Mud tracks, boundaries, and texture patterns made the wrong place look believable.

Classical features found **similar structure**, not the same place.

Same place, **different appearance**

The satellite looked greener and bushier — the UAV view looked barer and changed.

UAV QUERY

TOKEN 833



satellite: greener / bushier

UAV: **barer** / **changed land**

Same place — but would the algorithm know it?

Same place, **different appearance**

The satellite looked greener and bushier — the UAV view looked barer and changed.

UAV QUERY

TOKEN 833



satellite: greener / bushier

UAV: **barer** / **changed land**

CLASSICAL TOP-1

TOKEN 833



wrong region

Changed appearance confused the classical match.

Same place, different appearance

The satellite looked greener and bushier — the UAV view looked barer and changed.

UAV QUERY

TOKEN 833



satellite: greener / bushier

UAV: barer / changed land

CLASSICAL TOP-1

TOKEN 833



wrong region

DINOv2-VLAD

TOKEN 833



recovered the place

The appearance changed, but the place identity remained stable enough for learned retrieval.

The appearance changed, but the place identity remained.

Rotation helps sometimes — but **not always**

*A learned descriptor is **more tolerant** — but rotated and repeated layouts can still confuse retrieval.*

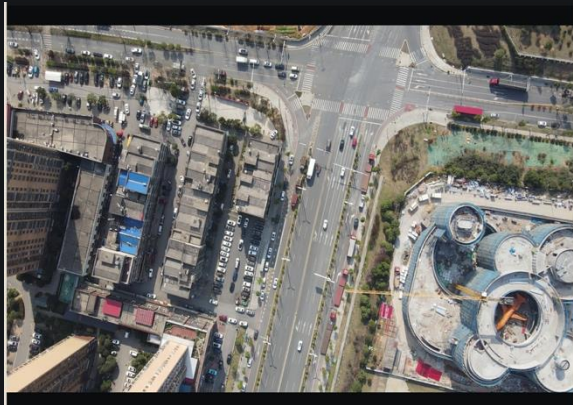
SOLVED UAV QUERY

UAV QUERY

TOKEN 591

DINO RECOVERY

TOKEN 591



same layout, **recovered** despite orientation shift

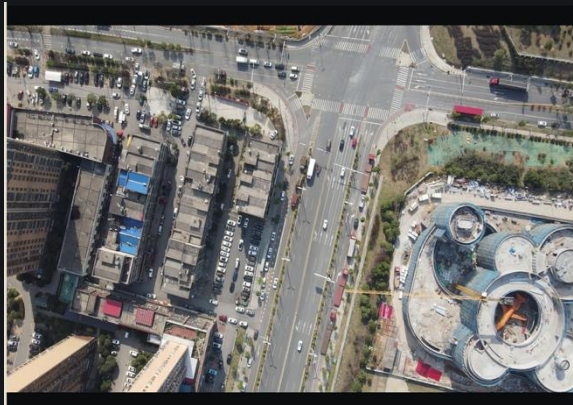
Some orientation shifts are **recoverable**.

Rotation helps sometimes — but **not always**

*A learned descriptor is **more tolerant** — but rotated and repeated layouts can still confuse retrieval.*

SOLVED UAV QUERY

UAV QUERY



TOKEN 591

DINO RECOVERY

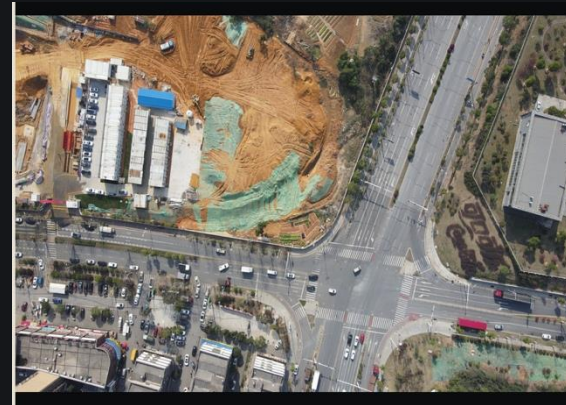


TOKEN 591

same layout, **recovered** despite orientation shift

UNSOLVED UAV QUERY

UAV QUERY



TOKEN 601

DINO CONFUSED



TOKEN 601

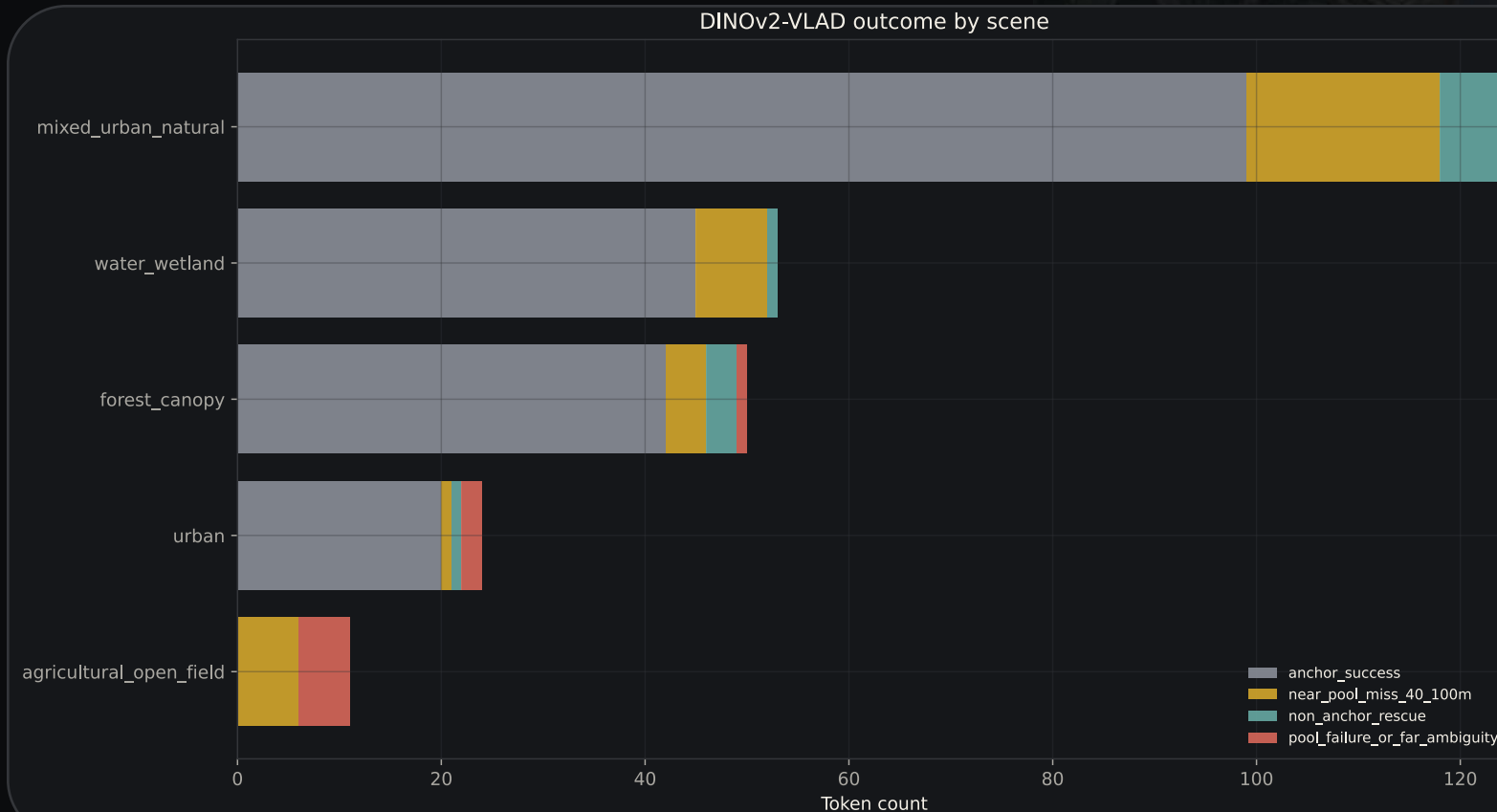
rotation plus repeated structure still creates ambiguity

Tolerance improved, but orientation is **not solved automatically**.

DINO helps, but **orientation remains open**.

DINO helped — but failures are **not random**

The next step is not only better retrieval — it is choosing the right strategy for each failure type.



FUTURE PLAN

Yaw-Aware cached Search

Rotate map candidates using drone yaw — or estimate rotation from visual motion.

3×3 map mosaics

Search neighborhoods, not only single tile centers.

Temporal voting

Use nearby frames before accepting a map correction.

DINO **changed the failure mode** — not the whole problem.